

SFT Pipeline — Data Distribution Report

Run directory: /scratch/project_462000963/users/aralikatte/sft-pipeline/runs/research_stage3_001

Exported at: 2026-04-28T13:25:12.068391+00:00

Total prompts: 14,879,293

Stages present: stage3, stage1

Domains: 9

Languages: 30

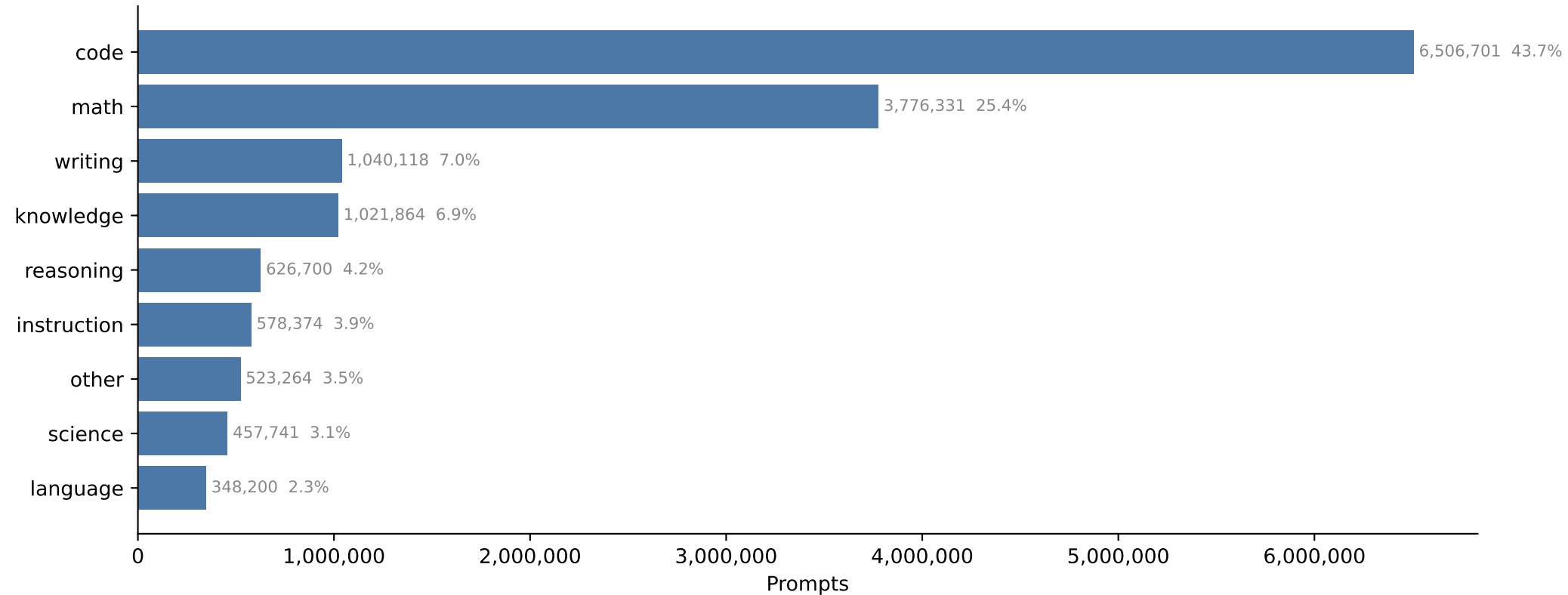
Sources: 30

Topics: 100

Domain Distribution

Each prompt is annotated with a high-level domain (e.g. math, code, science) by the LLM annotator. This chart shows how many prompts fall into each domain across the full dataset. A healthy distribution avoids extreme skew — Stage 4 quota sampling will rebalance this, so large imbalances here are expected and fine. Domains with very few prompts may be merged into 'other' at sampling time.

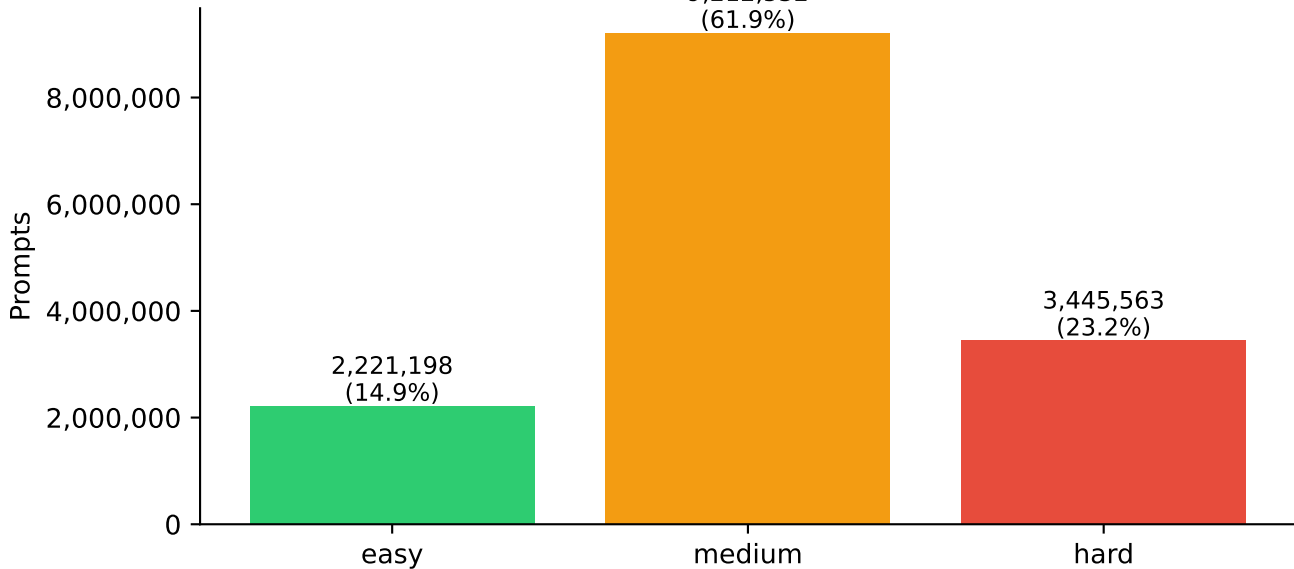
Domain Distribution (N=14,879,293)



Difficulty Distribution

Difficulty (easy / medium / hard) is assigned by the LLM annotator based on the complexity of reasoning required. Ideally the dataset skews toward medium and hard to maximise learning signal. A large 'easy' fraction means many prompts require only shallow reasoning; Stage 4 quota sampling can correct this by down-weighting easy and up-weighting hard prompts.

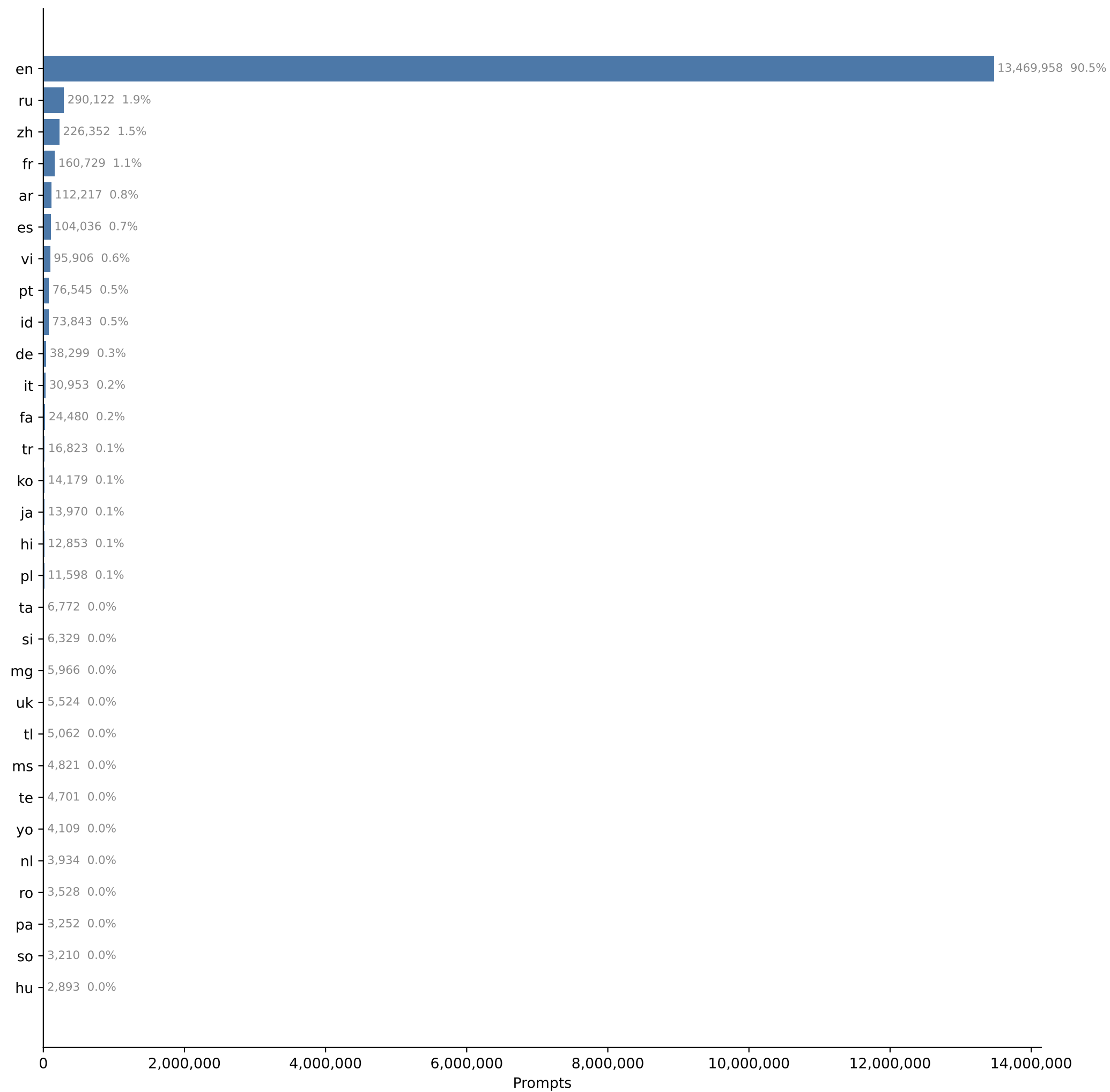
Difficulty Distribution



Language Distribution

Language is detected by the LLM annotator from the prompt text. Most source datasets are English-dominant, so 'en' will usually be the overwhelming majority. Non-English prompts come primarily from multilingual sources (e.g. OPUS, CC-100 derivatives). This chart shows the top 30 languages by raw count. **If Stage 7 translation is enabled in v2, these non-English prompts will be translated to English before fine-tuning.**

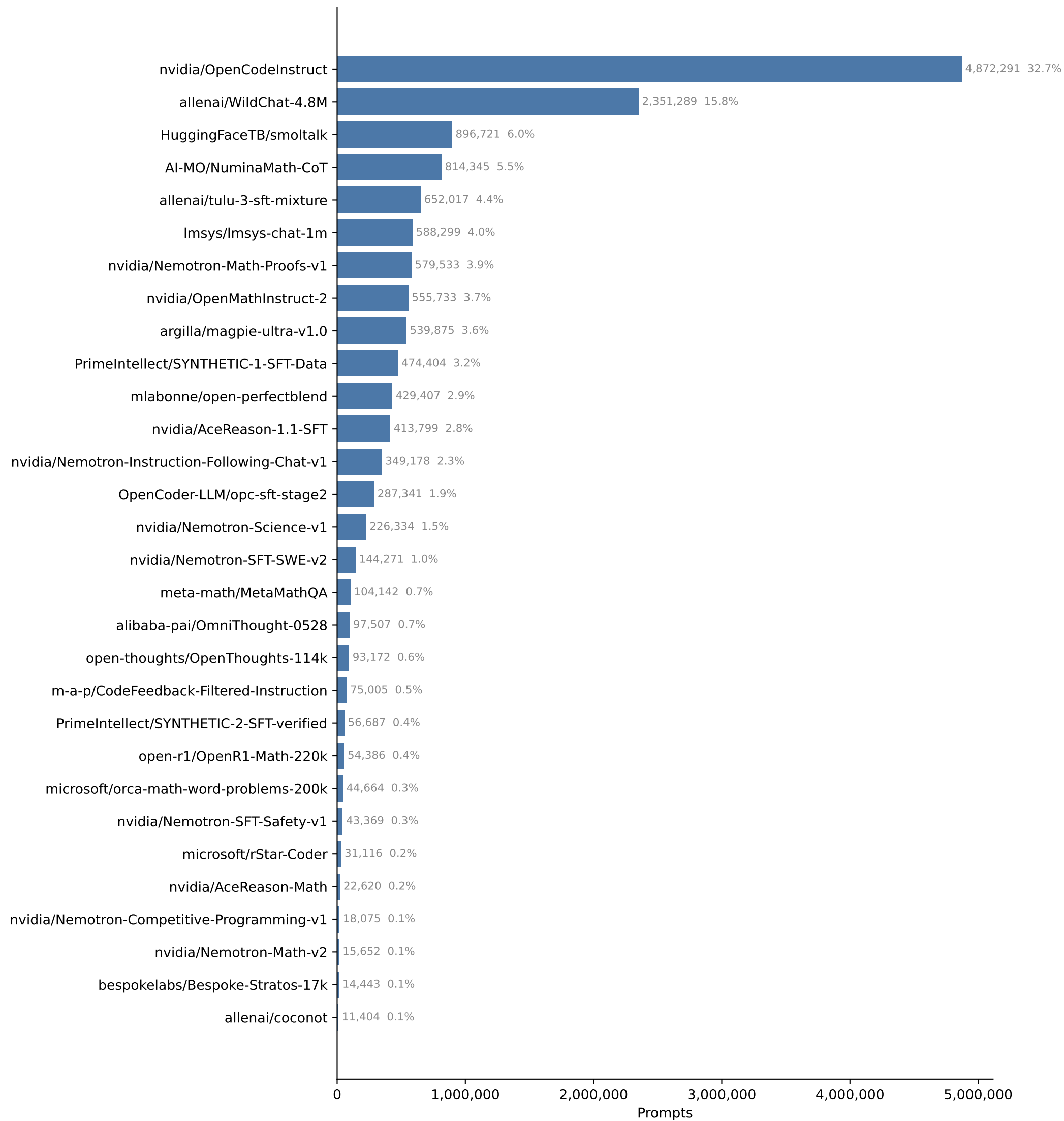
Language Distribution (top 30)



Source Dataset Distribution

Shows which HuggingFace datasets (or local sources) contributed how many prompts after Stage 1 collection and exact deduplication. Large sources dominate here; near-duplicate removal in Stage 4 will further reduce the effective contribution of sources with repetitive content. Use this chart to spot sources with unexpectedly low yield (possible download or parsing issues) or sources that are disproportionately large.

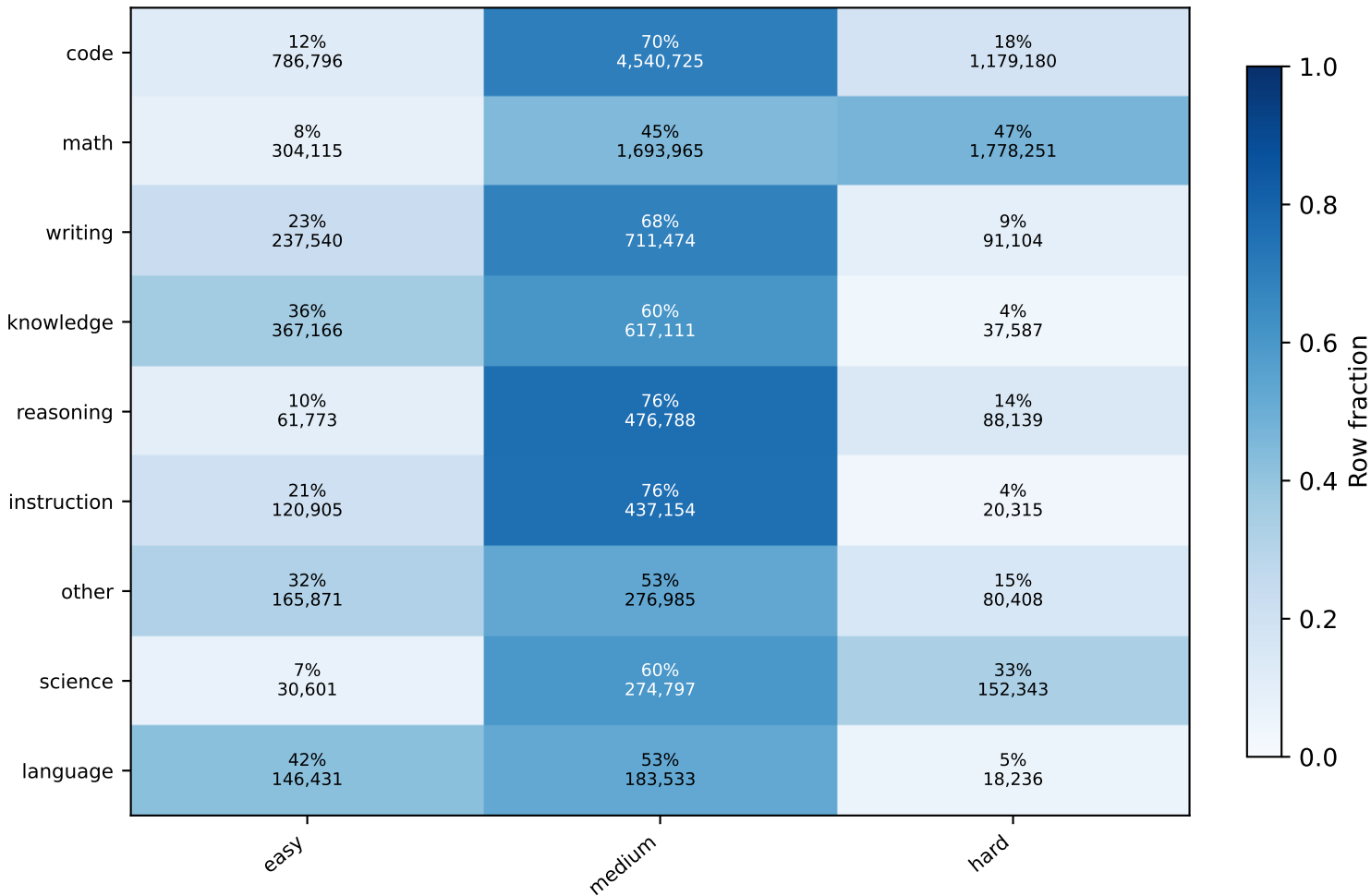
Source Dataset Distribution (top 30)



Domain × Difficulty Heatmap

Cross-tabulation of domain and difficulty, row-normalised (each row sums to 100%). Each cell shows the percentage AND absolute count for that (domain, difficulty) pair. Darker blue = higher fraction. Use this to spot domains where the LLM annotator systematically assigned one difficulty — e.g. if 'math' is 90% hard, that may reflect genuine data characteristics or annotator bias. These distributions directly inform the quota targets you should set in stage4_sample.yaml.

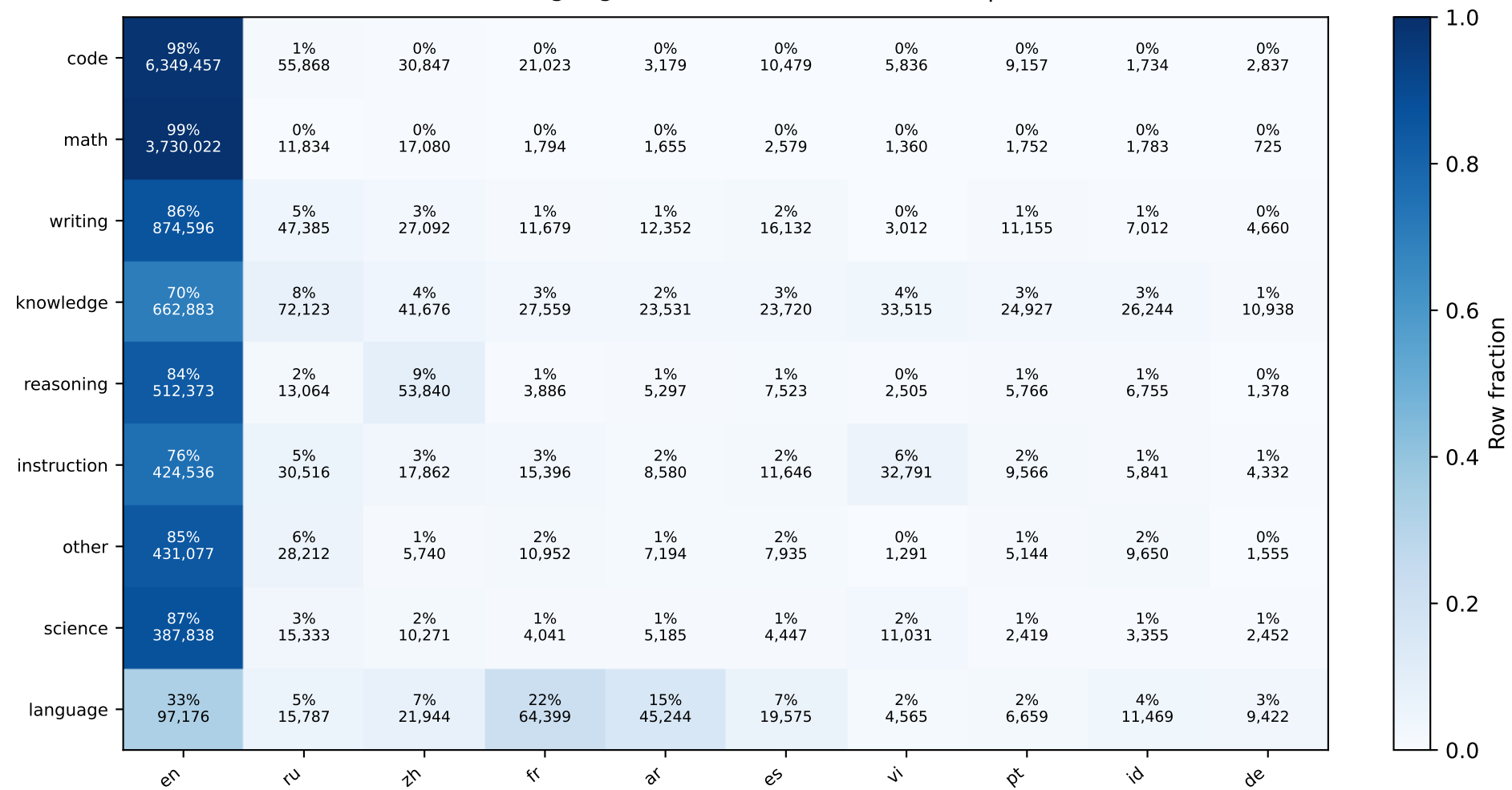
Domain × Difficulty (row-normalised; cell = % | count)



Domain × Language Heatmap

Cross-tabulation of domain and language (top 10 languages), row-normalised. Shows whether non-English prompts are evenly spread across domains or concentrated in specific ones. For example, coding prompts are often English-only while math word problems may appear in many languages. Cells with high non-English fraction in a domain suggest that domain's quota may need adjustment if the fine-tuning target is English-only.

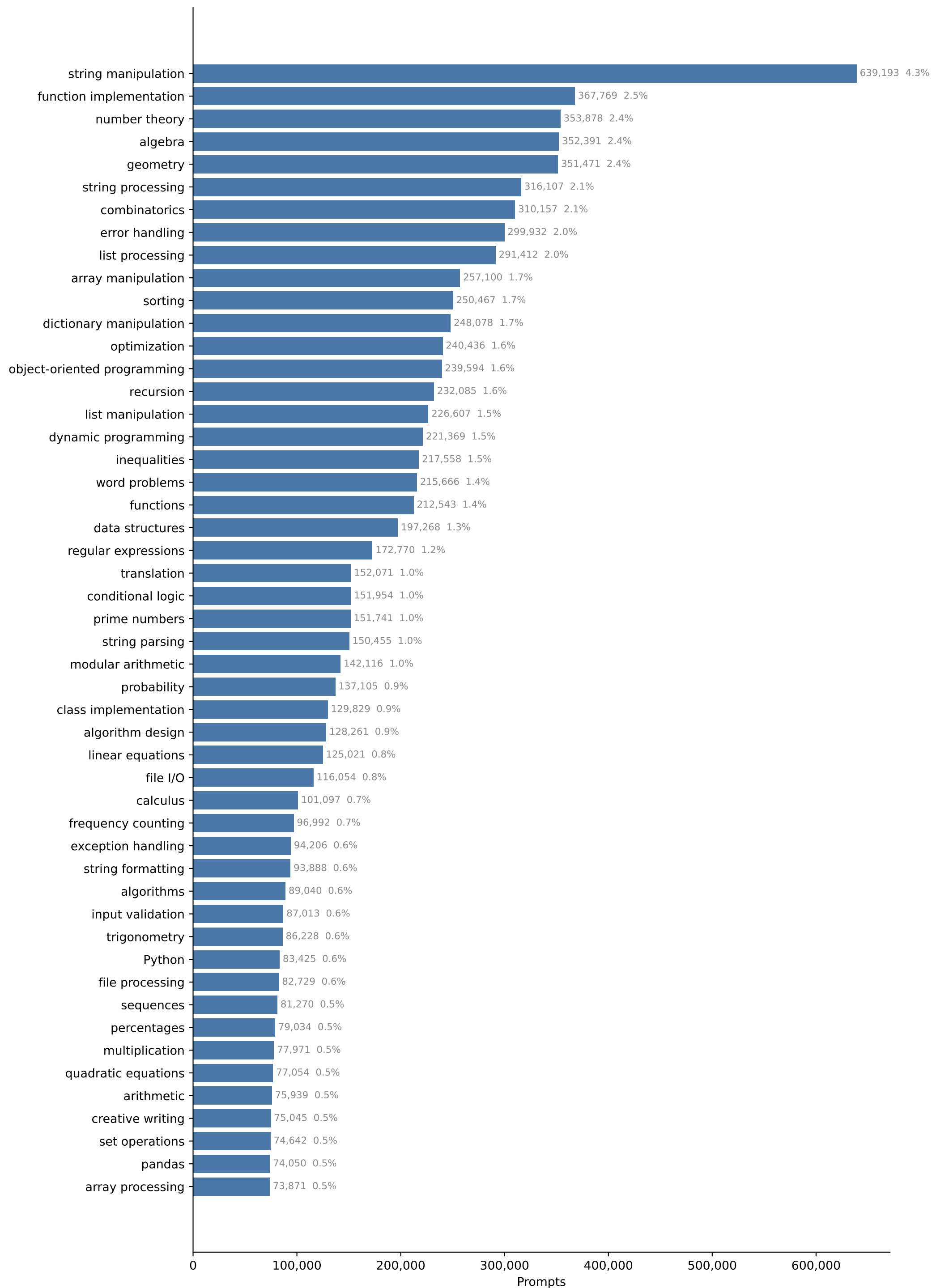
Domain × Language (row-normalised; cell = % | count)



Topics

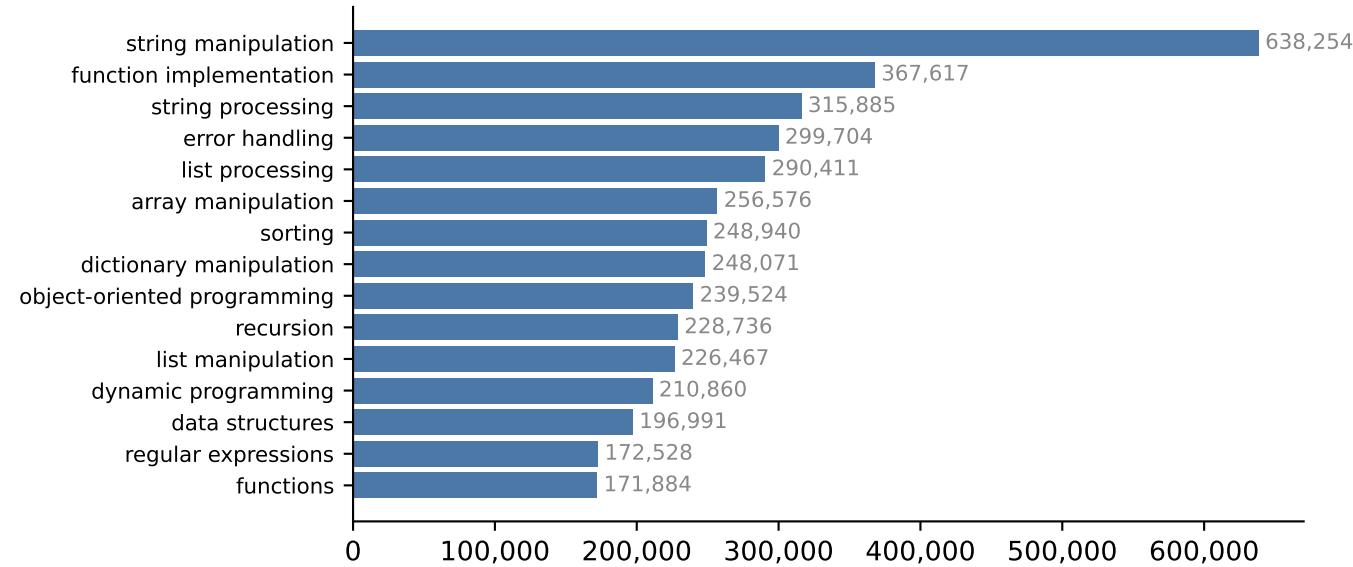
Each prompt is tagged with 1-5 fine-grained topic strings by the LLM annotator (e.g. 'linear algebra', 'sorting algorithms', 'organic chemistry'). Topics are stored as a list and exploded for counting — a single prompt contributes one count per topic it is tagged with. The overall chart shows the 50 most frequent topics across all domains. The per-domain pages that follow show the top 15 topics within each domain, which is more actionable for identifying coverage gaps.

Top 50 Topics (overall)

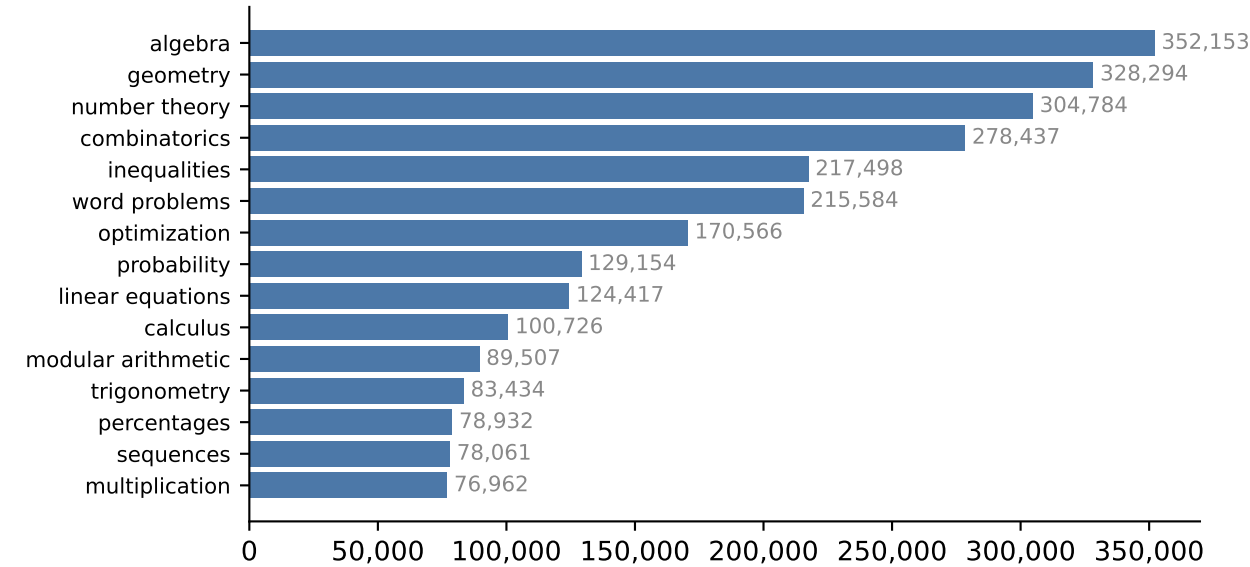


Top Topics per Domain

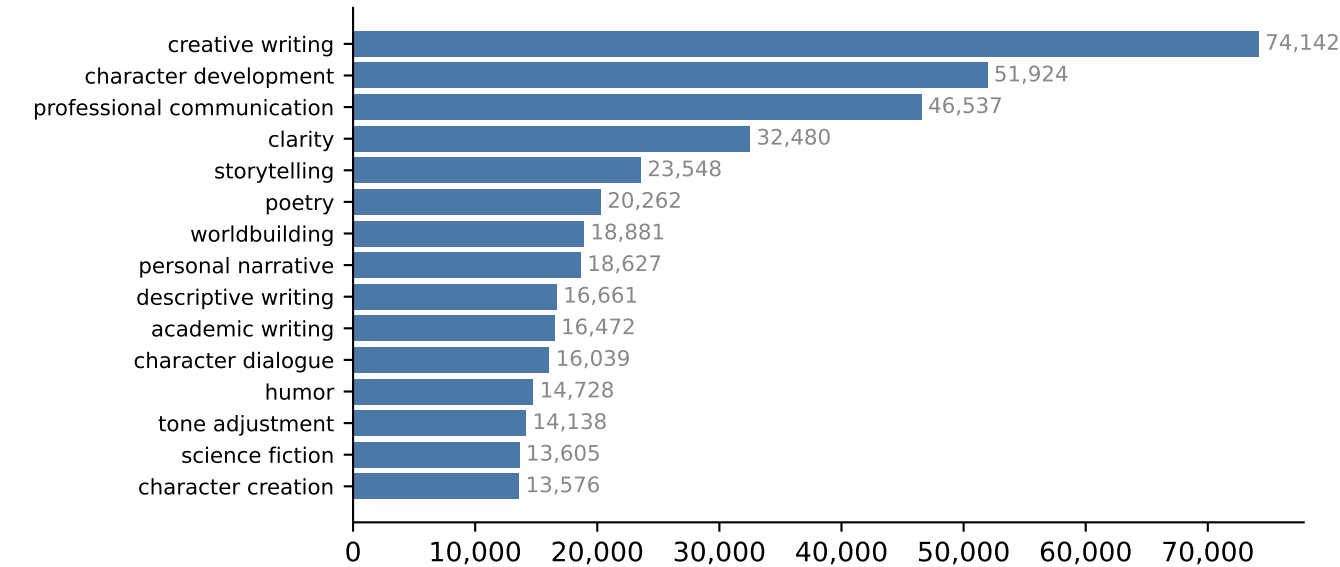
code (6,506,701 prompts)



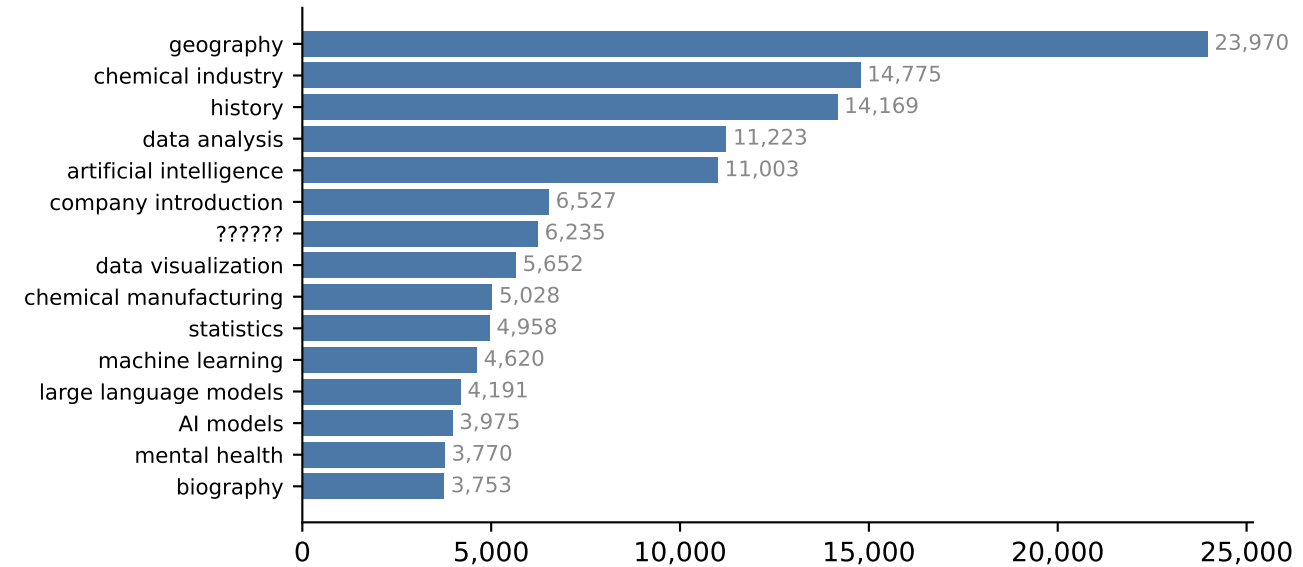
math (3,776,331 prompts)



writing (1,040,118 prompts)

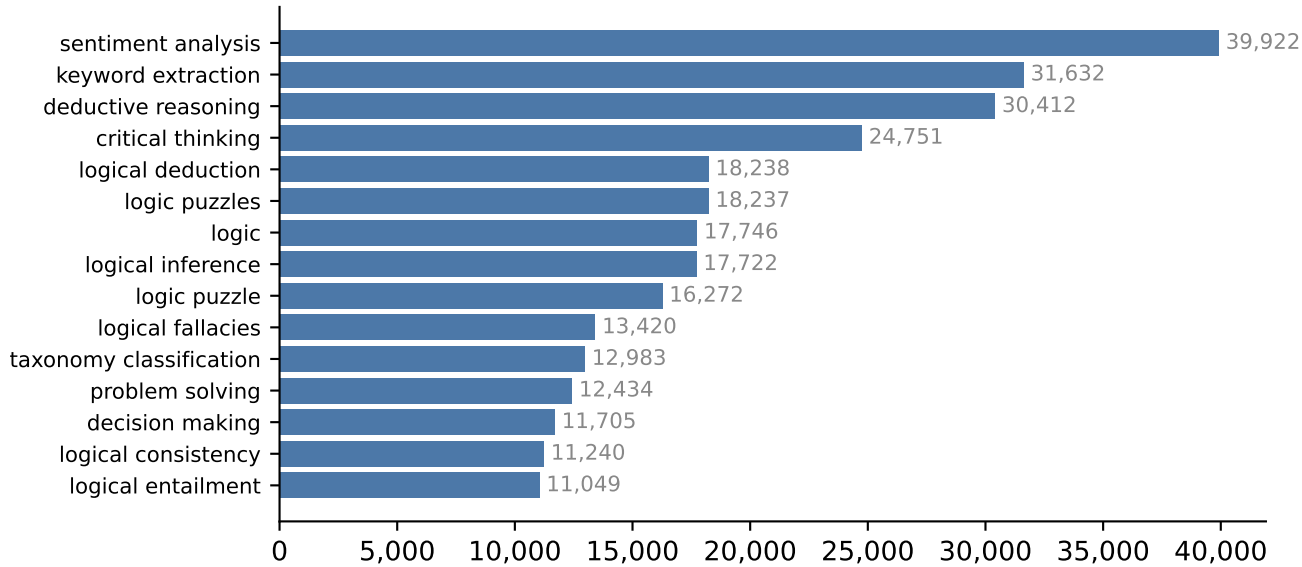


knowledge (1,021,864 prompts)

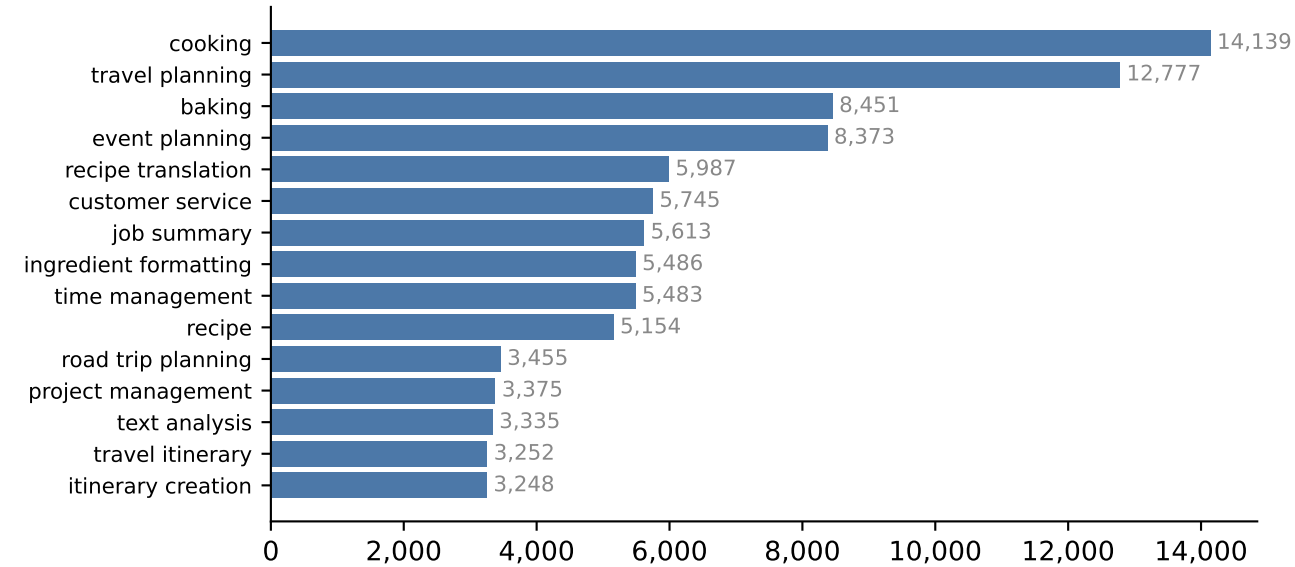


Top Topics per Domain

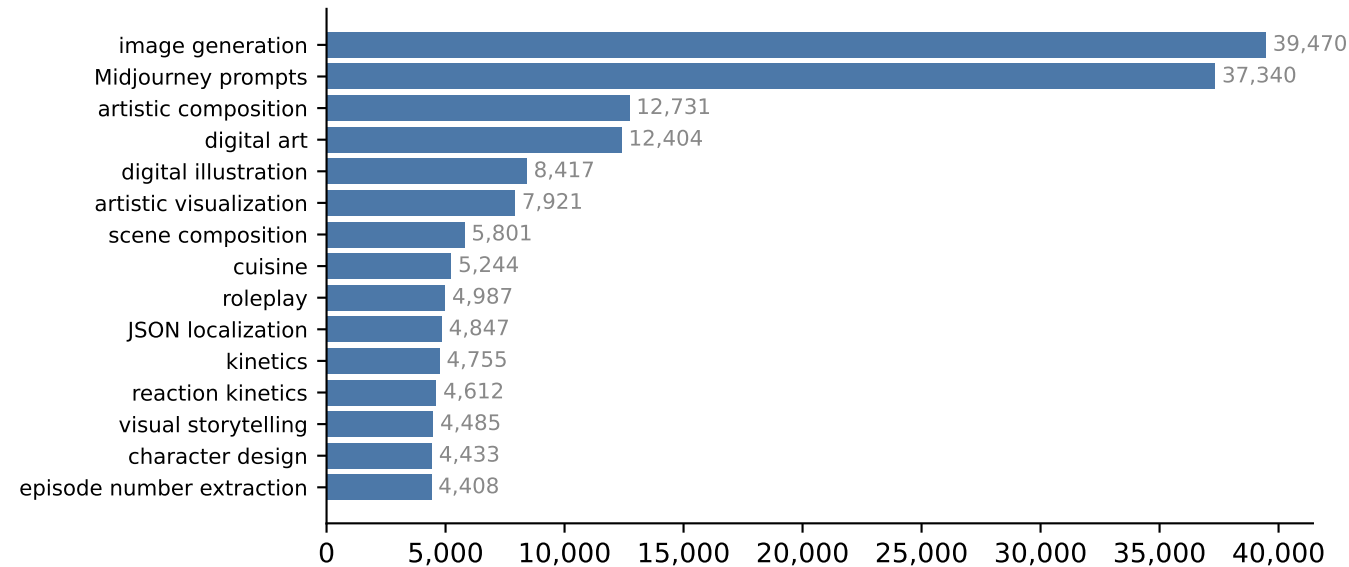
reasoning (626,700 prompts)



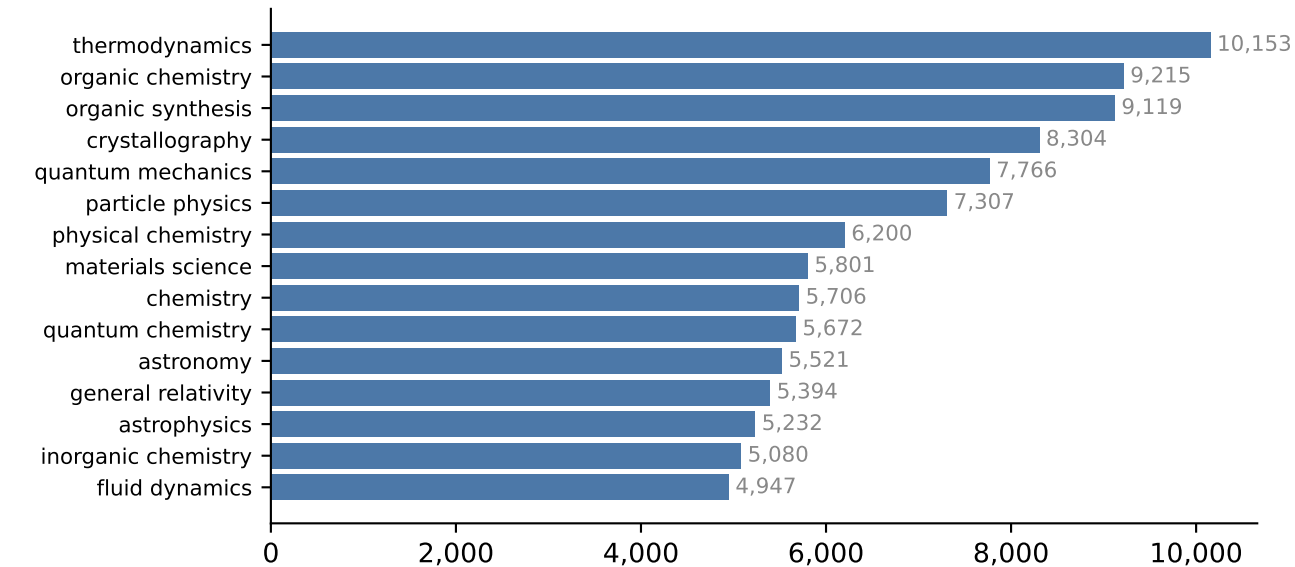
instruction (578,374 prompts)



other (523,264 prompts)

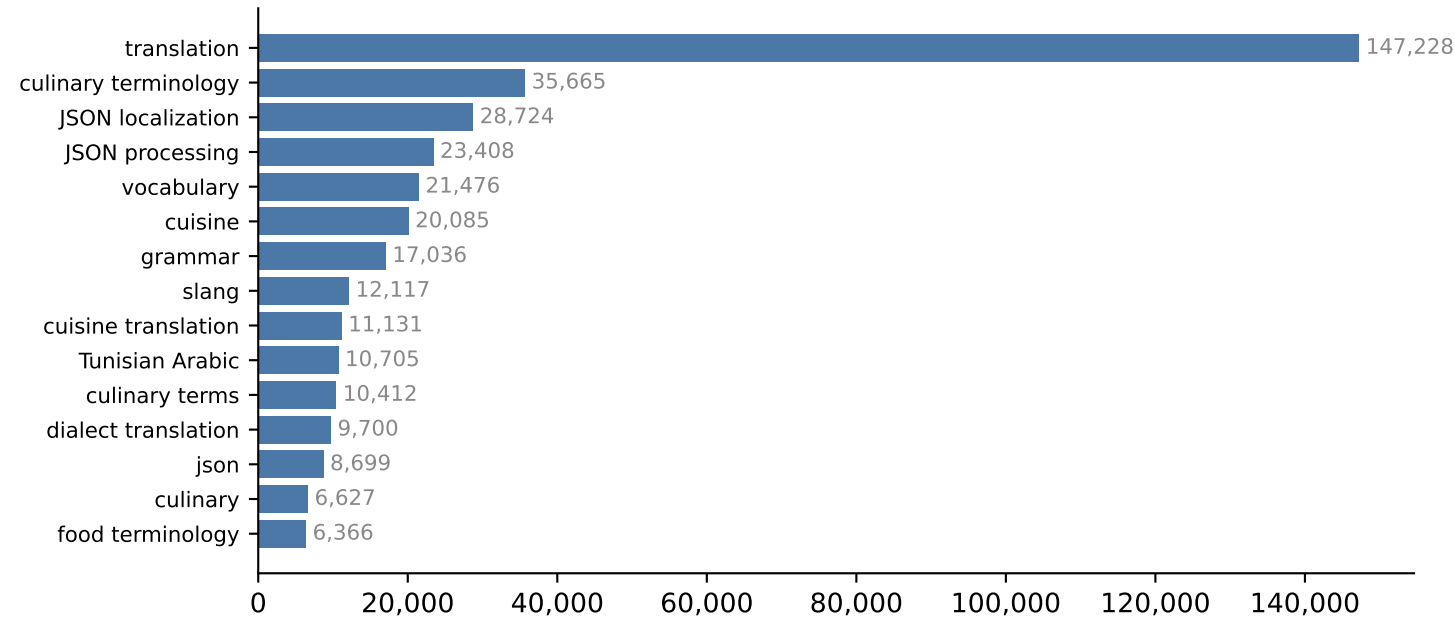


science (457,741 prompts)



Top Topics per Domain

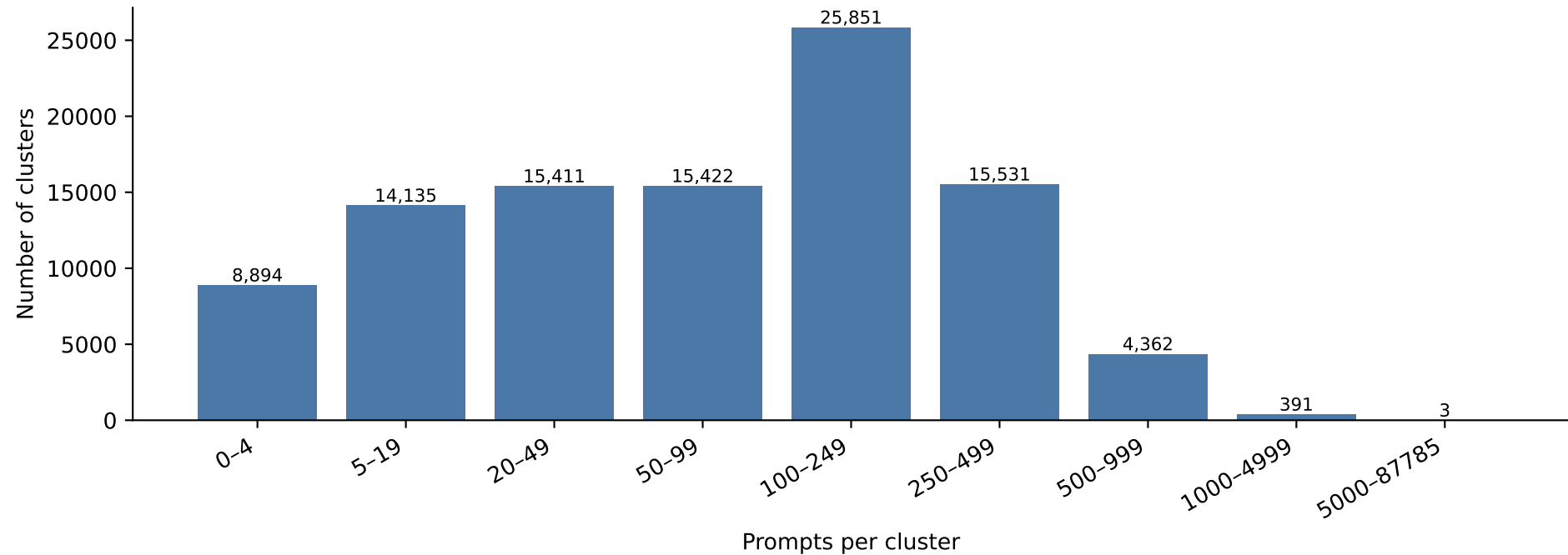
language (348,200 prompts)



Cluster Analysis

Stage 3 grouped all prompts into 100,000 clusters using GPU-accelerated K-Means on 1024-dimensional sentence embeddings. Clusters represent semantic neighbourhoods — prompts within a cluster ask about similar concepts. The size histogram shows how evenly sized the clusters are: many singleton or tiny clusters indicate sparse/noisy semantic regions; very large clusters suggest a topic is over-represented. Stage 4 near-dedup uses cosine similarity within clusters to remove near-duplicates, so cluster quality directly affects sample diversity. The 'clusters per domain' chart shows how many clusters are dominated by each domain.

Cluster Size Distribution (100,000 clusters)



Clusters Dominated per Domain

